

Care Delivery Organizations and Governments Should Take a Closer Look at Video in Healthcare

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This research outlines the opportunities, drivers and challenges for video in healthcare, and provides recommendations for CIOs at care delivery organizations (CDOs). CIOs should use this report to kick-start an evaluation of video and to avoid missing the opportunities it offers.

Key Findings

- The major opportunities for video in healthcare are for administrative meetings, clinician-to-clinician meetings, and video visits between clinicians and patients.
- Key areas of technical progress with video in healthcare include networks, reimbursement, high-definition video, technology convergence, security and desktop videoconferencing.
- The biggest challenges to video in healthcare are licensing, legal liability, credentialing, business justification and lack of leadership.

Recommendations

CDOs should:

- Focus on the economic benefits of using video for administrative purposes, rather than attempting to quantify the economic benefits of using video in patient care.
- Concentrate on training end users and changing processes and cultures.
- Watch the unified communications and collaboration (UCC) market, because it offers the potential of reduced costs, easier management and greater usability. In particular, the ability to combine desktop videoconferencing with real-time capture and manipulation of images is a highly promising development.
- Pay particular attention to the role of site coordinator for video visits.

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ANALYSIS

An increasing number of use cases, technical progress, and expanding availability of funding and reimbursement indicate a growing role for video in care delivery and administration. Now is the time for CDOs and governments to take a closer look at using video in healthcare.

Video offers several potential benefits in healthcare:

- Better use of clinicians' time
- More-rapid diagnosis and intervention
- Improved access to care, resulting in increased patient satisfaction and retention
- More patients kept in the local facility, keeping revenue from ancillary services such as laboratory tests in the local community
- A reduction in travel costs for clinicians, CDOs and patients, resulting in reduced carbon emissions

1.0 How Is Video Being Used in Healthcare?

Despite the undoubted benefits of video for patient care, CDOs will inevitably find it easier to quantify the benefits of using video for administrative purposes, rather than attempting to quantify its benefits for patient care.

The main use cases for video are:

- **Less travel for internal administrative meetings.** In large, multihospital organizations such as integrated delivery networks in the U.S. or National Health Service (NHS) trusts and health boards in the U.K., internal meetings consume large amounts of time and money.
- **Medical education.** Many medical education courses can be conducted via video, creating an opportunity for academic medical centers to provide branded video courses to other CDOs to generate revenue and attract referrals.
- **Clinician-to-clinician team meetings.** This is especially important for medical specialties such as cancer, where the patient is under the care of multiple clinicians at multiple locations, and there is a need for those physicians jointly to discuss images and treatment programs. The NHSs of England, Wales and Scotland make extensive use of video for such meetings.
- **Clinician-to-clinician consultations.** These typically link CDOs that lack specialist expertise in certain areas to other CDOs that can provide it. This is especially relevant for rural or community hospitals, and for providing supplementary emergency room coverage. Examples are:
 - Telestroke in the emergency department. In 80% of stroke cases, damage from strokes can be reduced if patients are given thrombolysis drugs, but they must be administered within three hours of the onset of symptoms, and frequently a neurologist is not available in the required time frame. A remote neurologist can view the patient over a videoconference and determine eligibility for thrombolysis. Specialist On Call and Reach Call are two examples of private companies that

deliver telestroke services to emergency departments via video. NHS Scotland is setting up a telestroke service that would cover all hospitals in Scotland.

- Pediatric ultrasound for rural hospitals that need an immediate opinion on what course of action to pursue for a difficult pregnancy. Examples of CDOs providing such advice are the University of Arkansas for Medical Sciences, CareGroup in Boston, Bon Secours Richmond Health System in Virginia, and Royal Hospital for Sick Children (Yorkhill) in Glasgow.
- "Telepharmacy," where a licensed pharmacist at a central site supervises a pharmacy technician who prepares a prescription at a remote site.
- Telementoring of resident doctors in rural hospitals, who can benefit from obtaining support through video links to their peers and mentors in the academic medical center where they trained.
- The growing adoption of video for clinician-to-clinician consultations is evident in the recent announcement by Asklepios, a 95-hospital CDO that is one of the largest healthcare providers in Germany. The CDO will install Web-based videoconferencing equipment on the 13,000 PCs of all its physicians by the end of 2008 to enable clinician-to-clinician consultations.
- **Patient-to-family communications.** This is relevant for patients who are in hospitals far from their homes, schools or other sources of support.
- **Interpretation services.** These are relevant for patients who require communication in foreign languages or sign language.
- **Rounding robots.** These are remote-controlled mobile robots equipped with cameras, computer screens and microphones that permit physicians to see and communicate with patients in hospital. Physicians, patients and CDOs have found value in their early uses of these robots. Basic kinks in the technology and its deployment have been worked out at sites in the U.S. and Canada, and it is in use at some mainstream community health systems, not only at academic medical centers. Because of technical progress, adoption is likely to expand gradually in terms of the number of organizations, as well as the number of applications of the technology per organization. See "Hype Cycle for Healthcare Provider Technologies and Standards, 2008."
- **Video visits.** These are prescheduled meetings between patients and clinicians that are conducted using video. Video visits are best suited for medical disciplines where the caregiver normally does not need to touch the patient, and their value is multiplied when patient transportation is difficult due to distance, patient frailty or security (for example, for prison inmates). Video visits are growing in use in healthcare, especially in mental health and rehabilitation. Indeed, some mental health professionals in the U.S. exclusively practice through video visits. However, it is hard to track the total number of video visits, because there is no Current Procedural Terminology (CPT) code that identifies whether a visit was conducted using video. Among government-funded health systems, the U.S. Department of Veterans Affairs uses video visits extensively for mental health and rehabilitation consultations (it treated 43,000 patients using video visits in 2007 and expects this to climb to 74,000 by the end of 2008). Other large users of video visits in the U.S. are the Alaska Federal Healthcare Access Network (AFHCAN), the U.S. Indian Health Service, the University of California Davis Health System, the University of Arizona and the University of Texas Medical Branch, which provides video visits for the majority of the prisons of Texas. In Canada, the efforts of the Infoway program have led to extensive usage of video visits across the country. In

Europe, Denmark uses video visits extensively for mental health. Usage of video visits in other countries is mostly limited to pilot projects at present. A critical success factor for video visits is the role of the site coordinator. That person sets up the consultations and makes sure the equipment works, the patient is correctly positioned and the room is appropriately lit. The site coordinator also is responsible for promoting the use of video visits to patients and clinicians.

2.0 What Factors Are Driving Progress in the Use of Video in Healthcare?

- **Networks.** The cost and availability of high-speed network connections have been the principal impediments to video. In some European countries, including Norway, Sweden and the U.K., governments have established national high-speed networks reserved for healthcare use. In the U.S., two federal government organizations provide subsidies to states and rural CDOs for setting up high-speed network connections: the Universal Service Administrative Company and the Federal Communications Commission (FCC). The FCC is spending more than \$400 million on 69 programs reaching more than 6,000 healthcare facilities. Such programs, combined with grants from state governments, are enabling hospitals to establish regional telemedicine networks. Instead of the "hub and spoke" model for video visits, with a single tertiary-care facility with point-to-point connections to community hospitals, a growing number of U.S. rural telemedicine networks comprise multiple tertiary-care facilities that compete for video visits and remote diagnoses from multiple community hospitals. "Hub" CDOs benefit by gaining business for their employed physicians. "Endpoint" CDOs benefit by keeping patients in their local facilities. The move toward a network-based model is being facilitated by the transition of video traffic from ISDN (which requires a point-to-point connection) to IP (any-to-any connection). Because these networks are dedicated for healthcare use, they are better able to guarantee quality of service than if the video traffic were routed over the regular Internet.
- **Reimbursement.** One factor encouraging the adoption of video visits in the U.S. is the increased availability of reimbursement. The U.S. Medicare program reimburses both the "originating" (or referring) site and the "distant" (or treating) site for video visits, including office visits, mental health consultations, and pharmacology consultations. As of 1 January 2009, skilled nursing facilities, in-hospital dialysis centers and community mental health centers will become eligible originating sites for Medicare reimbursement, in addition to existing eligible sites (hospitals, physician offices, rural health clinics and federally qualified health centers). This expansion in Medicare reimbursement, which resulted from the recent approval in Congress of H.R. 3661, will undoubtedly stimulate greater use of video in healthcare. Medicaid reimbursement varies by state, as does the degree to which state legislation mandates private payer reimbursement. In most government-funded health systems, reimbursement is less of an issue than in the U.S., because there is a tighter link between the provider and the payer of healthcare. However, the move in some European countries, such as Germany, England and France, to paying hospitals per procedure will complicate this situation and will require governments to define reimbursement procedures for video visits.
- **High-definition video.** While there is much hype about "telepresence" — the combination of high-definition (HD) video with meeting rooms that are specially designed to replicate a face-to-face meeting — CDOs can get benefits similar to those from telepresence at about a tenth of the price by replacing their standard-definition (SD) video cameras with HD cameras. HD offers significantly better image quality than SD, and the price of HD is falling rapidly. It will open the door for more extensive use of video

to view images remotely, and it will enhance disciplines such as stroke care where the ability to view facial features is especially important. HD bandwidth requirements vary by manufacturer; for example, the University of California Davis uses a minimum of 1MB/second per channel for HD, compared with 384KB/second for SD. For sessions supporting HD video plus HD content-sharing devices, such as laptops and high-resolution document cameras, Gartner recommends a minimum of 2MB/second per channel. Vendors of HD cameras include LifeSize, Polycom, Radvision and Tandberg. For a detailed discussion of HD, see "High-Definition Videoconferencing Moves to the Forefront."

- **Improvements in interoperability.** The leading videoconferencing vendors have enhanced their multiconference units (MCUs) in recent years. These provide the ability to combine different resolutions from different endpoints in the same call, without forcing users to downgrade to the lowest resolution. This means that, if there are two HD participants and one SD participant on a call, the HD participants will see each other in HD.
- **Technology convergence.** Video visits are being enhanced by UCC — the aggregation of diverse communication and collaboration tools, including voice, audio/Web/videoconferencing, social software and e-mail. Video vendors are looking to enhance the functionality of popular desktop communication applications, such as Microsoft Live Communications Server/Office Communications Server and IBM Lotus Sametime, through the integration of videoconferencing. With this capability, a vendor's video system will be able to initiate a videoconference call (including HD) via a click-to-call feature from an IM session. Added to this is the increased ability of medical devices to connect to video terminals. For example, a technician assisting a patient at a rural clinic could use a specially enabled Web camera to capture an image of the patient's body part, and a remotely located physician could zoom in or out on the image while on a desktop videoconference. While end users benefit from having a variety of communication options at their disposal, IT managers benefit from the economies of scale provided by a shared common infrastructure.
- **Improvements in security.** Due to the growing use of VPNs, an increasing proportion of video traffic is encrypted. In addition, most videoconferencing codecs now include built-in encryption. Video traffic must be encrypted to meet the requirements of the U.S. Health Insurance Portability and Accountability Act (HIPAA).
- **Emergence of desktop and remote-worker videoconferencing.** Until now, most video has used dedicated rooms at physical facilities. Desktop videoconferencing is being facilitated by server-based videoconferencing solutions, by the integration of video with IP telephony, and by features such as call forwarding and message forwarding/playback that more closely replicate the capabilities of voice communications and improve ease of use. Connecting to remote workers at home offices is being facilitated by firewall transversal software, which enables remote devices known to a CDO to communicate across a firewall. Theoretically, this could also facilitate communication with mobile devices, but this raises issues such as coverage, quality of service, screen size and delivering video to a device that users are holding to their ears.

3.0 What Are the Main Challenges Preventing More-Widespread Deployment of Video in Healthcare?

- **Licensing.** In the U.S., physicians receive separate licenses from each state in which they wish to practice medicine. U.S. state licensing bodies vary greatly in the extent to

which they allow physicians licensed in other states to treat state residents using video. One reason why the U.S. Department of Veterans Affairs has been able to use video so extensively is that, as a federal organization, it is exempted from state licensing requirements. In Europe, each country has its own licensing arrangements, and organizations that wish to provide or receive video-based treatment across national boundaries must make ad hoc agreements. Because of these problems, the use of video in healthcare is most common within states and countries rather than across borders.

- **Legal liability.** Although confusion remains regarding which jurisdiction has authority over cross-border care delivery, there appears to be growing consensus that, if a clinician in state or country A provides telemedicine services to a patient in state or country B, the laws of B should apply. In the U.S., liability insurance regulations, such as caps on damages and statutes of limitation, vary widely from state to state, and outside the U.S., each country has its own laws relative to healthcare. This is another reason why cross-border health delivery is rare.
- **Credentialing.** The Joint Commission in the U.S., a national quality accreditation body, specifies that practitioners who treat patients via telemedicine must be credentialed with the organization that receives the telemedicine service. These standards apply to "treating" physicians (those who diagnose, treat or prescribe), not to physicians providing "consulting" or "interpretive" services. This is skewing the use of video toward second opinions rather than direct patient care.
- **Business justification.** Early deployments of video have been plagued by problems such as poor image quality, lack of end-user training on the equipment, lack of attention to changing clinical processes, and a view that video is futuristic and "fun" rather than a must-have. Therefore, while many CDOs have purchased videoconferencing equipment, few use it to its full potential. To make the most of video in healthcare, CDOs will need to focus on end-user training, cultural change, and clear demonstrations of the business value of video. The move to HD will help; organizations that have switched from SD to HD cameras have experienced significant increases in usage rates.
- **Lack of leadership.** A related problem has been the lack of executive buy-in for the use of video. Leaders who have grown up in health systems that emphasize face-to-face communication may struggle to see the value of video. This applies to individual CDOs as well as to governments. Even in government-dominated health systems such as in the U.K. and the Nordic countries, where there is a close alignment between provider and payer and therefore an incentive to adopt innovations such as video that can reduce costs and improve care quality, governments have not yet taken much action to promote the use of video in healthcare.

4.0 Recommendations

- CDOs should focus on the economic benefits of using video for administrative purposes, rather than attempting to quantify the economic benefits of using video in patient care.
- CDOs that want to make the most of video in healthcare will need to focus on training end users and changing processes and cultures.
- CDOs should look for government or private programs that offer assistance with connecting to high-speed broadband networks, and with the setup and ongoing costs of installing videoconferencing equipment.

- CDOs should keep a close watch on the UCC market, because it offers the potential of reduced costs, easier management and greater usability. In particular, the ability to combine desktop videoconferencing with real-time capture and manipulation of images is a highly promising development.
- CDOs should closely track the evolution of the videoconferencing market. Stand-alone videoconference meeting rooms that use dedicated ISDN lines are being replaced with IP-based, high-definition videoconferencing solutions, telepresence technology, or lower-bandwidth, webcam, multiparty solutions.
- CDOs should pay particular attention to the role of site coordinator for video visits.
- To facilitate clinician adoption, CDOs should ensure that video visits fit into existing referral patterns.
- CDOs should ensure physicians know that, at any time, they can request an in-person visit if they do not believe a video visit is appropriate.
- CDOs should block out time on physicians' schedules for video visits. They should not expect physicians to do them "after hours."
- Video visits should be part of a "patient of the future" strategy that includes home and mobile monitoring, e-visits, remote image readings and personal health records. CDOs should recognize that desktop video technology will soon be advanced enough for patients to have video visits through their PCs from their homes or workplaces. This will be coupled with advances in the medical device market that will allow patients to connect inexpensive peripherals to their computers and share data on blood pressure, weight and medical images in real time with clinicians. This will potentially open up an opportunity for third-party companies to sell video visits directly to patients, competing with brick-and-mortar CDOs.

RECOMMENDED READING

"Hype Cycle for Telemedicine, 2008"

"Case Study: A Texas Telemedicine Program Offers Lessons for Governments and Care Delivery Organizations Worldwide"

"High-Definition Videoconferencing Moves to the Forefront"

"Hype Cycle for Enterprise Communication Applications, 2008"

"MarketScope for Video Telepresence Solutions, 2008"

"Key Issues for Unified Communications and Collaboration, 2008"

"Cost Cutting Options for IP Videoconferencing"

"The Gartner View on Enterprise Video"

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